

Japan's Semiconductor Renaissance

Rapidus, TSMC Kumamoto, and the Race to Reclaim Chipmaking Sovereignty

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Executive Summary

Japan's semiconductor industry is attempting one of the most ambitious industrial restoration projects in modern economic history: rebuilding advanced chip manufacturing capabilities that were lost over three decades of strategic drift. The twin engines of this restoration are Rapidus — a government-backed consortium targeting 2nm logic production by 2027 — and TSMC's Kumamoto facility (JASM), which pivoted from 16nm to 3nm production in a 2025 strategic upgrade that transformed the facility from a cost-reduction exercise into a genuine technology asset for Japan's automotive and industrial electronics industries. Japan's government has backed this effort with extraordinary financial commitment: approximately JPY 10 trillion (\$67B) in semiconductor and digital industry investment over the period 2022-2030, with JPY 150 billion injected directly into Rapidus in 2025 alone.

Japan's semiconductor restoration is not merely a commercial endeavour — it is a national security project. The Japanese government has explicitly framed semiconductor manufacturing capability as critical national infrastructure, drawing lessons from the COVID pandemic's supply chain disruptions and the US-China semiconductor war's implications for nations without domestic advanced chip production. Whether Rapidus can actually deliver 2nm production chips by 2027 — a timeline that most semiconductor industry analysts regard as aggressive — is the defining question for Japan's semiconductor decade.

Part I — Japan's Semiconductor Fall: Three Decades of Managed Decline

The Heights From Which Japan Fell

In 1988, Japanese companies produced approximately 50% of the world's semiconductor chips by value. NEC, Hitachi, Toshiba, Fujitsu, and Mitsubishi dominated DRAM, leading edge logic, and memory chips. The government-orchestrated VLSI (Very Large Scale Integration) research consortium of 1976-1979 had produced breakthrough manufacturing technologies in photolithography and defect reduction that briefly made Japan the global leader in semiconductor process technology.

The fall from these heights has been well-documented: US trade sanctions (1986 US-Japan Semiconductor Agreement capping Japanese market share), the rise of Korean DRAM producers (Samsung, SK Hynix) with lower cost structures and aggressive capital deployment, and Taiwan's foundry model (TSMC, 1987) disrupting the vertically integrated IDM approach that Japanese companies depended on. By 2000, Japan's global semiconductor market share had fallen to approximately 20%; by

2020, it had fallen to approximately 8-10%.

Japan retained specific niches — semiconductor manufacturing equipment (TEL, Nikon, Advantest, Shin-Etsu Chemical), semiconductor materials (photoresists, CMP slurries, polishing pads), and specialty chips for automotive and industrial applications. These niches are genuine — Japan controls approximately 50% of global semiconductor materials supply and 30% of global semiconductor equipment supply by value — but they represent supporting industries, not the manufacturing leadership Japan once held.

Part II — Rapidus: The Moon Shot Bet

The Architecture of Ambition

Rapidus was established in August 2022 as a privately-held joint venture among eight Japanese companies: Toyota, Sony, NTT, NEC, SoftBank, Denso, Kioxia, and Mitsubishi UFJ Financial Group. The company's stated mission is to produce 2nm-class logic chips in Japan by 2027, in partnership with IBM's technology licensing and Belgium's imec research consortium. The government's backing is explicit: the Japanese government (through METI's Industrial Innovation Organisation, NEDO) has provided and committed approximately JPY 920 billion (\$6.2B) in total through 2025-2026, with additional funding tranches expected through 2027.

The scale of Rapidus's ambition is genuinely extraordinary. No company outside of TSMC, Samsung, and Intel has successfully manufactured chips at 2nm-class process nodes. TSMC spent approximately \$40B over more than a decade developing the process technology and manufacturing infrastructure for its N2 node (currently beginning volume production in Taiwan). Rapidus proposes to replicate and operate a comparable process node in Chitose, Hokkaido with a fraction of that investment and timeline — its current facility investment of approximately \$10-15B is 25-35% of what TSMC spent on N2 development alone.

Industry scepticism is widespread. ASML, the Dutch lithography monopolist whose EUV machines are essential for 2nm production, has supplied Rapidus with an initial EUV system for R&D. The engineering leap from EUV-equipped research to volume EUV manufacturing — requiring not only the tools but the process chemistry, defect metrology, yield learning, and supply chain management that TSMC has spent decades developing — cannot be accomplished in 5 years by a startup, in the view of most semiconductor veterans.

The Revised Ambition: 2027 as Pilot, Not Volume

Reading between Japan's government communiques and Rapidus management statements, the 2027 target should be understood as "pilot production" or "customer sampling" rather than "volume manufacturing." This is the distinction TSMC itself makes between early node development (running customer test wafers at low yield to validate process parameters) and high-volume manufacturing (HVM, running at 80%+ yield for major production orders).

Rapidus's pilot production line at Chitose should be operational in 2027, accepting customer test designs. Volume production — at the yields and throughputs that make 2nm economically viable for commercial fabless customers — is more realistically targeted for 2028-2029. The Japanese government appears to understand this timeline internally, even if public communications have maintained the 2027 framing.

The IBM-imec Technology Foundation

Rapidus's technology foundation rests on IBM's 2nm chip architecture (announced May 2021 — the first demonstration of 2nm GAAFET nanosheet transistors) and imec's process integration expertise. IBM's 2nm is a research demonstration, not a production process; the Rapidus-IBM partnership involves

translating IBM's demonstration into a manufacturable process flow. Imec, which hosts process development partnerships with TSMC, Intel, Samsung, and essentially every major semiconductor company, provides Rapidus access to leading-edge process research that would otherwise take decades to develop internally.

The critical question is whether IBM research process + imec integration expertise + Rapidus manufacturing scale-up can compress a development cycle that took TSMC approximately 10-15 years (from research to HVM) into a 5-year programme. The answer determines whether Rapidus is a genuine semiconductor renaissance or an expensive industrial policy exercise.

Part III — TSMC Kumamoto: The Anchor That Changed its Nature

From 16nm to 3nm: The Strategic Pivot

TSMC's Japan Advanced Semiconductor Manufacturing (JASM) facility in Kumamoto, Kyushu, was originally announced in 2021 as a 12nm/16nm facility — mature nodes by TSMC standards, but still representing a significant improvement over Japan's existing domestic semiconductor capability. The facility received approximately ¥476 billion (\$3.2B) in Japanese government subsidies.

In 2025, JASM announced a strategic upgrade: a second phase of the Kumamoto facility would produce 3nm-class chips (using TSMC's N3E or equivalent process), with additional government subsidy of approximately JPY 300 billion. This pivot transforms JASM from a "cost-reduction exercise in a friendly democracy" into a genuine advanced technology facility relevant to Japan's most sophisticated electronics applications.

The automotive and industrial electronics implications are profound. Modern automotive microcontrollers and AI inference chips for autonomous driving (ADAS Level 2-4) increasingly require 5nm-7nm class processes for the performance and power efficiency targets that Tier 1 automotive suppliers demand. Toyota, Denso, and Japanese automotive Tier 1s have been entirely dependent on TSMC in Taiwan for these chips — with the supply chain vulnerabilities made vividly apparent by the COVID semiconductor shortage of 2021-2022, which caused Toyota production cuts of approximately 400,000 vehicles. JASM Phase 2 at 3nm in Kumamoto provides Japan's automotive industry with domestic advanced chip supply security.

JASM Customer Base

JASM Phase 1 (mature nodes) customer base includes Sony Semiconductor Solutions (image sensors — Sony is one of the world's largest image sensor producers), Denso (automotive MCUs), and various Japanese electronic component makers. JASM Phase 2 (3nm) is expected to serve Toyota's automotive SoC requirements, Sony's next-generation AI image processing chips, and potentially NTT's custom network processor designs.

Part IV — Japan's Materials and Equipment Moat

The Silent Semiconductor Superpower

Japan's most enduring semiconductor competitive advantage is in manufacturing materials and equipment — the picks-and-shovels of the chip industry. The concentration of critical semiconductor supply in Japanese companies is remarkable:

Photoresists: JSR, Shin-Etsu Chemical, and Tokyo Ohka Kogyo collectively control approximately 75-80% of global advanced photoresist supply. ArF immersion and EUV photoresists — the materials that enable sub-7nm patterning — are effectively Japanese monopolies.

Silicon wafers: Shin-Etsu Handotai and Sumco together produce approximately 50% of global 300mm silicon wafer supply. Every chip made by TSMC, Samsung, Intel, and all other major manufacturers is made on silicon wafers that are predominantly Japanese in origin.

CMP Slurries and Polish Pads: Fujimi Incorporated, Fujibo Ehime, and Cabot Microelectronics Japan dominate the chemical mechanical planarisation consumables market, essential for multilayer chip fabrication.

Lithography Systems: Tokyo Electron (TEL) and Nikon each control critical lithography and process equipment segments. TEL is the world's largest semiconductor process equipment company by revenue after ASML and Applied Materials — providing etching, deposition, and cleaning systems essential across all technology nodes.

Japan's materials and equipment position is the semiconductor industry's most concentrated chokepoint outside of ASML's EUV monopoly. Any disruption to Japanese semiconductor materials supply — natural disaster, trade dispute, or geopolitical conflict — would cascade through the global chip industry within weeks. This concentration gives Japan extraordinary, if often underappreciated, leverage in semiconductor geopolitics.

Part V — The Geopolitical Dimension: Japan-US-Netherlands Semiconductor Alliance

Export Controls: Japan's New Strategic Role

Japan's decision in 2023 to align with US semiconductor export controls — restricting the export of certain semiconductor manufacturing equipment to China — represents the most significant Japanese security policy shift in decades outside of the defense rearmament discussion. Japanese equipment (particularly TEL's etch and deposition systems) had been a critical input for China's semiconductor industry; the export control alignment with the US creates a material constraint on China's fab expansion plans.

This alignment with US export controls is not costless: China is a major market for Japanese semiconductor equipment companies. TEL's China revenue fell approximately 30% year-on-year in 2024 following the export controls; the company has been redirecting capacity toward non-China customer growth. However, Japan's government judged that the geopolitical alignment benefits — deeper technology cooperation with the US, Netherlands (ASML), and other allied nations — outweigh the short-term commercial losses.

Part VI — Strategic Outlook: Scenarios for Japan's Semiconductor Decade

Scenario A: Rapidus Achieves Pilot by 2028, Customer Wins Follow (Probability: 35%)

Rapidus achieves commercial pilot production of 2nm-class chips by 2027-2028, attracts one or two Japan-domestic anchor customers (automotive SoC, high-performance computing for AI), and secures additional government funding for scale-up to 20,000 wafer starts/month by 2030. JASM Phase 2 at 3nm begins production 2027, cementing Japan's position as a genuine multi-node semiconductor hub. The Japan semiconductor restoration thesis gains global credibility; additional foreign FDI in Japan's semiconductor ecosystem follows.

Scenario B: Rapidus as Research Facility, JASM as Commercial Anchor (Probability: 50%)

Rapidus achieves technical demonstrations at 2nm but fails to secure volume commercial orders due to yields and cost below competitive levels. The facility transitions into a national research asset (a well-equipped advanced process lab for Japan's universities and corporate R&D) rather than a commercial foundry. JASM Phase 2 at 3nm becomes Japan's primary advanced manufacturing anchor — commercially credible because of TSMC's process heritage and yield management capability. Japan has limited domestic advanced logic manufacturing but a world-class R&D base and materials/equipment position.

Scenario C: Rapidus Timeline Failure, Programme Restructure (Probability: 15%)

Rapidus misses the 2027 pilot target significantly (beyond 18 months), triggering political review of the programme's cost-benefit and potential restructuring. The programme costs ¥5+ trillion before producing commercial value, creating fiscal controversy. Japan's semiconductor strategy reorients toward materials/equipment excellence and JASM anchor rather than domestic advanced logic manufacturing.

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All data from METI, TSMC JASM disclosures, industry trackers, and cited news sources.